

Forms

Chapter-9

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Overview of Forms

- A form is an HTML element that contains and organizes **objects** called **form controls**, including text boxes, check boxes, and buttons, that can accept information from website visitors.

Two components of a form:

1. The **HTML form** itself, which is the web page user interface
2. The **server-side processing**, which works with the form data and sends e-mail, writes to a text file, updates a database, or performs some other type of processing on the server

Form Element

- The **form element** contains a form on a web page.
- The **<form>** tag specifies the beginning of a form area.
- The closing **</form>** tag specifies the end of a form area.
- Multiple forms on a web page, but cannot be **nested inside** one another.
- The form element can be configured with attributes that specify which server-side program or file will process the form, how the form information will be sent to the server, and the name of the form.

Cont'd

Attributes of the form element

Attribute	Value
action	URL of file name/path of server-side processing script
autocomplete	on (or) off
id	Alphanumeric, no spaces ; the value must be unique and not used for other id values on the same web page document
method	get (or) post
name	Alphanumeric, no spaces , begin with a letter; choose a form name value that is descriptive but short (for example, OrderForm is better than Form1 or WidgetsRUsOrderForm)

Form Controls

- The purpose of a form is to **gather information** from a web page visitor.
- Form controls are the **objects** that accept the information.
- Types of form controls include text boxes, scrolling text boxes, select lists, radio buttons, check boxes, and buttons.
- HTML5 offers new form controls, including those that are customized for e-mail addresses, URLs, dates, times, numbers, and even date selection.

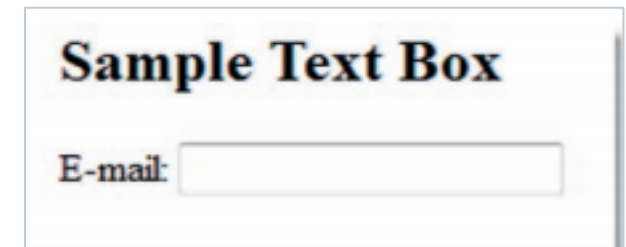
Input Element Form Controls

- The **input element** is a **stand-alone**, or tag that is used to configure several different types of form controls.
- The input element is not coded as a pair of opening and closing tags.
- Use the type attribute to specify the type of form control that the browser should display.

Text Box

- The **<input>** tag with **type="text"** configures a **text box** form control.
- The text box form control accepts **text** or **numeric information** such as names, e-mail addresses, phone numbers, and other text.

The code for the text box is shown below:



E-mail: `<input type="text" name="email" id="email">`

Cont'd

Common text box attributes

Attribute	Value
type	text
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
size	Numeric value
maxlength	Numeric value

Cont'd

Common text box attributes

value	Text or numeric character
disabled	disabled
autocomplete	on (or) off
required	required
accesskey	Keyboard character
tabindex	Numeric value

Submit Button

- The **submit button** form control is used to submit the form.
- The web server will invoke the server-side processing program or script listed on the **form's action** property.
- The input element with type="submit" configures a submit button.

For example,

```
<input type="submit">
```

Reset Button

- The **reset button** form control is used to reset the form fields to their **initial values**.
- A reset button does not submit the form.

For example,

```
<input type="reset">
```

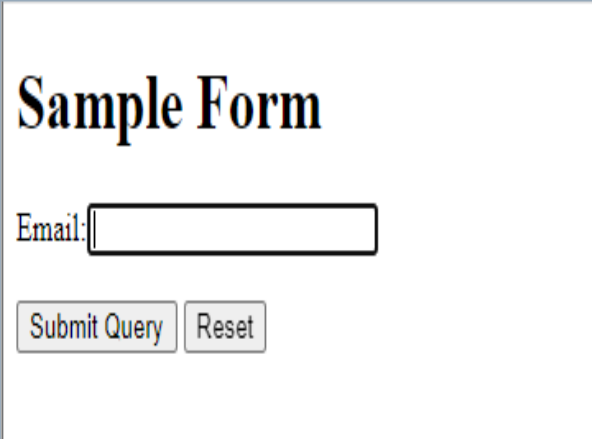
Cont'd

Common attributes for submit and reset buttons

Attribute	Value
type	submit (or) reset
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
value	Text or Numeric characters
accesskey	Keyboard character
tabindex	Numeric value

Sample Code

```
<body>
  <h1>Sample Form</h1>
  <form method="get">
    Email:<input type="text" Name="email" required="required" autocomplete="off">
      <br><br>
    <input type="submit" Name="submit" value="Submit Query">
    <input type="reset">
  </form>
</body>
```



Sample Form

Email:

Check Box

- The **check box** form control allows the user to **select one or more** of a group of predetermined items.
- The input element with **type="checkbox"** configures a check box.

Common check box attributes

Attribute	Value
type	checkbox
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
checked	checked
value	Text or numeric character

Cont'd

Common check box attributes

disabled	disabled
readonly	readonly
autofocus	autofocus
Required	required
accesskey	Keyboard character
tabindex	Numeric value

Sample Code

```
<h1>Sample check Box</h1>
```

```
<form method="get">
```

```
    Choose the browsers you use: <br>
```

```
    <input type="checkbox" name="IE" id="IE" value="yes"> Internet Explorer<br>
```

```
    <input type="checkbox" name="Firefox" id="Firefox" value="yes"> Firefox<br>
```

```
    <input type="checkbox" name="Opera" id="Opera" value="yes"> Opera<br>
```

```
</form>
```



Sample Check Box

Choose the browsers you use:

- ☐ Internet Explorer
- ☐ Firefox
- ☐ Opera

Radio Button

- The **radio button** form control allows the user to select **exactly one** (and **only one**) choice from a group of predetermined items.
- The input element with **type="radio"** configures a radio button.
- Each radio button in a group is given the same **name** attribute and a unique **value** attribute.

Cont'd

Common radio button attributes

Attribute	Value
type	radio
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
checked	checked
value	Text or numeric character

Cont'd

Common radio button attributes

disabled	disabled
readonly	readonly
autofocus	autofocus
Required	required
accesskey	Keyboard character
tabindex	Numeric value

Sample Code

```
<h1>Sample Radio Button</h1>
```

```
<form method="get">
```

```
Select your favorite browser:<br>
```

```
<input type="radio" name="favbrowser" id="favIE" value="IE"> Internet Explorer<br>
```

```
<input type="radio" name="favbrowser" id="favFirefox" value="Firefox"> Firefox<br>
```

```
<input type="radio" name="favbrowser" id="favOpera" value="Opera"> Opera<br>
```

```
</form>
```



Sample Radio Button

Select your favorite browser:

- ☐ Internet Explorer
- ☐ Firefox
- ☐ Opera

Hidden Input Control

- The **hidden input control** stores **text or numeric information**, but it is **not visible** in the browser viewport.
- Hidden controls can be accessed by both client-side scripting and server-side processing.
- The input element with **type="hidden"** configures a hidden input control.

For Example;

```
<input type="hidden" name="sendto" id="sendto" value="order@site.com">
```

Cont'd

Common hidden input control attributes

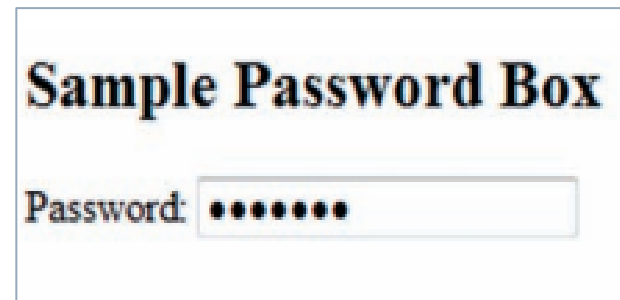
Attribute	Value
type	checkbox
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
value	Text or numeric character
disabled	disabled

Password Box

- The **password box** form control is **similar to the text box**, but it is used to **accept information** that must be **hidden** as it is entered, such as a password.
- The input element with type="password" configures a password box.

For Example;

Password: <input type="password" name="pword" id="pword">



A sample password box form. It features a title "Sample Password Box" in a bold, black, serif font. Below the title, the label "Password:" is followed by a text input field. The input field contains seven black dots, indicating that the password is hidden.

Scrolling Text Box

- The **scrolling text box** form control accepts **free-form comments, questions, or descriptions**.
- The **textarea element** configures a scrolling text box.
- The **<textarea>** tag denotes the beginning of the scrolling text box.
- The closing **</textarea>** tag denotes the end of the scrolling text box.

Cont'd

Common scrolling text box attributes

Attribute	Value
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
cols	Numeric value
rows	Numeric value
maxlength	disabled
tabindex	Numeric value

Cont'd

Common scrolling text box attributes

disabled	Numeric value
readonly	readonly
autofocus	autofocus
placeholder	Text or numeric characters
required	required
wrap	Hard or soft
accesskey	Keyboard character

Sample Code

```
<h1>Contact Us</h1>
```

```
<form method="get">
```

```
First Name:<input type="text" name="fname"
id="fname" required><br><br>
```

```
Last Name:<input type="text" name="lname"
id="lname"><br><br>
```

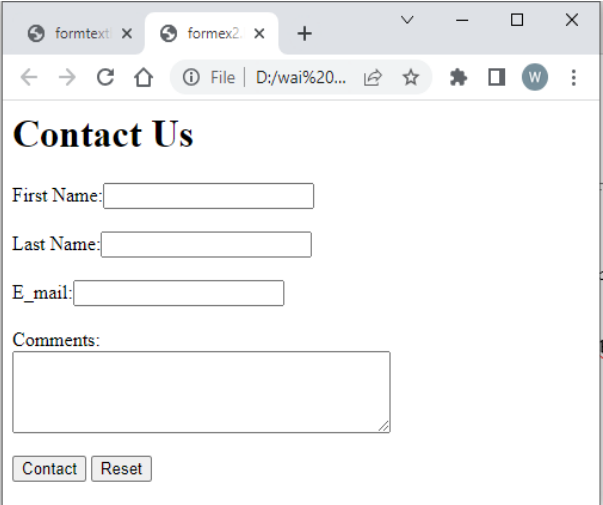
```
E_mail:<input type="text" name="email"
id="email" required><br><br>
```

```
Comments:<br>
```

```
<textarea name="comments" id="comments"
rows="4" cols="40" wrap="soft"
maxlength="20"></textarea><br><br>
```

```
<input type="submit" value="Contact">
```

```
<input type="reset"></form>
```

A screenshot of a web browser window showing the rendered HTML code. The browser has two tabs: 'formtext1' and 'formex2'. The address bar shows 'File | D:/wai%20...'. The page title is 'Contact Us'. The form contains three text input fields labeled 'First Name:', 'Last Name:', and 'E_mail:'. Below these is a larger text area labeled 'Comments:'. At the bottom of the form are two buttons: 'Contact' and 'Reset'.

Select List

- The **select element** contains and configures the select list form control.
- The **<select>** tag denotes the beginning of the select list.
- The closing **</select>** tag denotes the end of the select list.
- The **option element** contains and configures an option item displayed in the select list form control.
- The **<option>** tag denotes the beginning of the option item.
- The closing **</option>** tag denotes the end of the option item.

Cont'd

Common select element attributes

Attribute	Value
name	Alphanumeric, no spaces, begins with a letter
Id	Alphanumeric, no spaces, begins with a letter
size	Numeric value
multiple	multiple
disable	disabled
tabindex	Numeric value

Cont'd

Common option element attributes

Attribute	Value
value	text or numeric characters
selected	selected
disable	disabled

Sample Code

```
<h1>Select List Example</h1>
```

```
<form>
```

```
  <select size="4" name="jumpmenu" id="jumpmenu">
```

```
    <option value="index.html">Home</option>
```

```
    <option value="products.html">Products</option>
```

```
    <option value="services.html">Services</option>
```

```
    <option value="about.html">About</option>
```

```
    <option value="contact.html">Contact</option>
```

```
  </select>
```

```
</form>
```

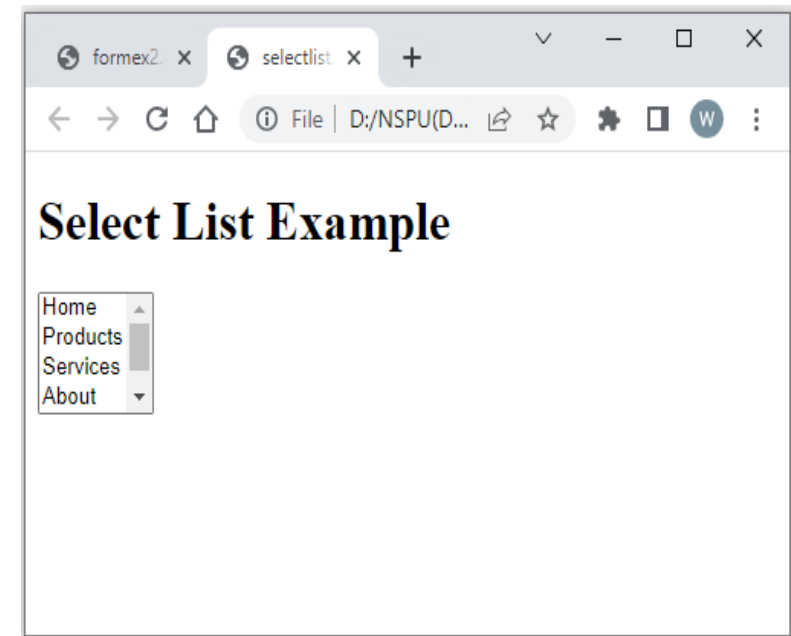


Image Buttons

- An image button is clicked or tapped, the form is submitted.
- The image button is coded using the `<input>` tag along with `type="image"` and a `src` attribute with the value of the name of the image file.

For example;

```
<input type="image" src="login.gif" alt="Login Button">
```

The Button Element

- Another way to add more interest to a form is to use the **button element**.
- Web page content that is between the **<button>** and **</button>** tags is configured to be part of the button.

Common button element attributes

Attribute	Value
type	Submit, reset or button
name	Alphanumeric, no spaces, begins with a letter
id	Alphanumeric, no spaces, begins with a letter
alt	Brief text description of the image
value	Text or numeric characters

Accessibility and Forms

- The accessibility of form controls, including the label element, fieldset element, legend element, tabindex attribute, and accesskey attribute.

label element

- The **label element** is a container tag that associates a text description with a form control.
- The **<label>** tag specifies the beginning of the label.
- The closing **</label>** tag specifies the end of the label.

Cont'd

➤ There are two different methods to associate a label with a form control.

1. The first method places the label element as **a container**

```
<label>E-mail: <input type="text" name="email" id="email"></label>
```

2. The second method uses the **for** attribute to associate the label

```
<label for="email">E-mail: </label>
```

```
<input type="text" name="email" id="email">
```

Notice that the value of the **for attribute** on the label element is the same as the value of the id attribute on the input element.

Fieldset and Legend Elements

- Form is to group elements of a similar purpose together using the **fieldset element**.
- The **legend element** provides a text description for the fieldset grouping.

Sample Code

```
<h1>Contact Us</h1>
```

```
<form>
```

```
<fieldset>
```

```
<legend>Customer Information</legend>
```

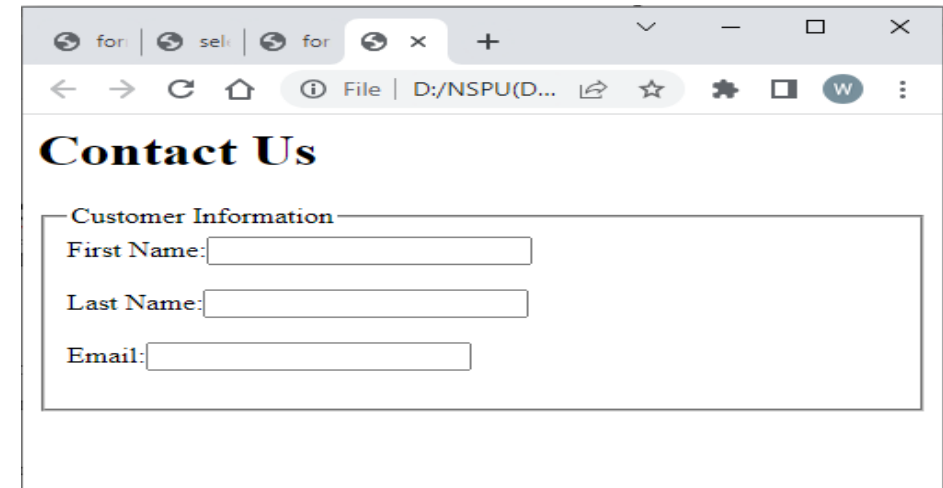
```
<label>First Name:<input type="text"
```

```
    name="fname" id="fname" required></label><br><br>
```

```
<label>Last Name:<input type="text" name="lname" id="lname"></label><br><br>
```

```
<label>Email:<input type="text" name="email" id="email" required></label><br><br>
```

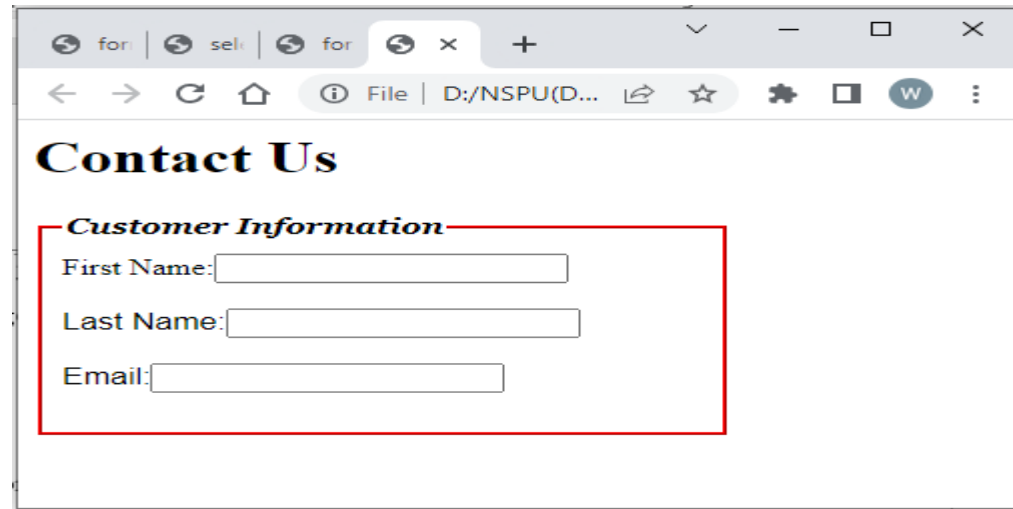
```
</fieldset> </form>
```



CSS Sample Code

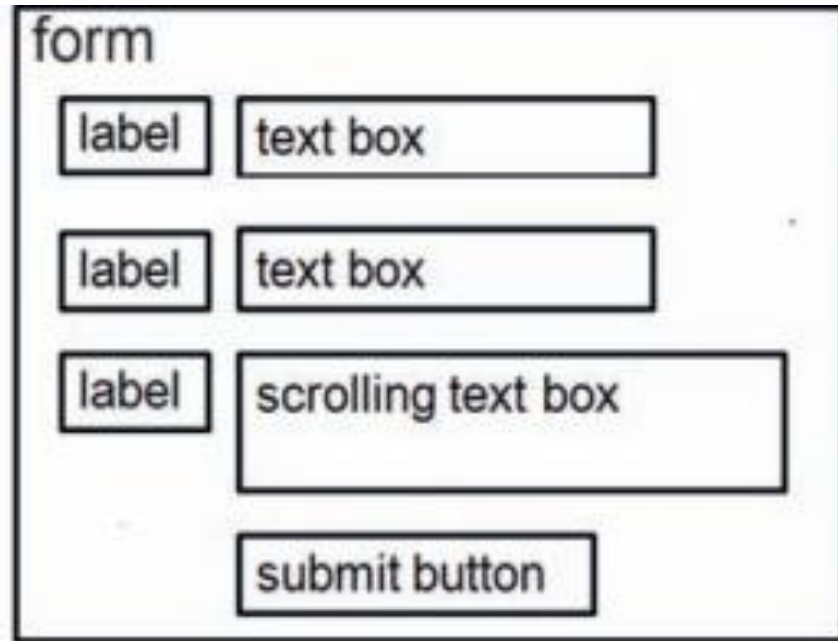
```
fieldset { width: 320px;  
border: 2px ridge #ff0000;  
padding: 10px;  
margin-bottom: 10px; }
```

```
legend {  
font-family: Georgia, "Times New Roman", serif;  
font-weight: bold;  
font-style: italic; }  
label { font-family: Arial, sans-serif; }
```

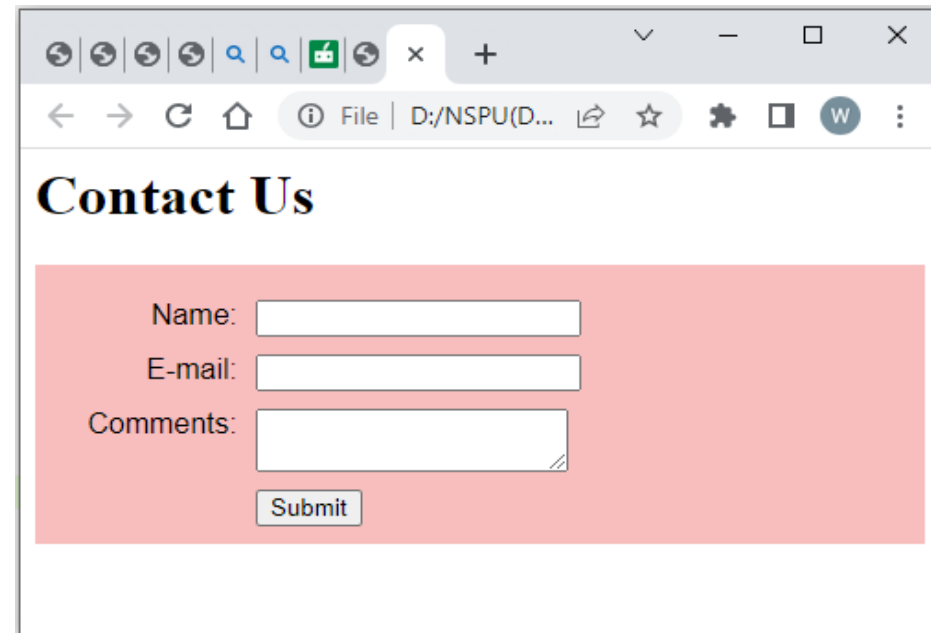


Style a Form with CSS

- A form with CSS, the box model is used to create a series of boxes.



Wireframe for a form



This form is configured with CSS

CSS Sample Code

```
form { background-color:#eaeaea;  
font-family: Arial, sans-serif;  
padding: 10px; }  
label { float: left;  
width: 100px;  
clear: left;  
text-align: right;  
padding-right: 10px;  
margin-top: 10px; }  
input, textarea { margin-top: 10px;  
display: block; }  
#mySubmit { margin-left: 110px; }
```

Sample Code

```
<h1>Contact Us</h1>
<form> <label for="myName">Name:</label>
<input type="text" name="myName" id="myName">
<label for="myEmail">E-mail:</label>
<input type="text" name="myEmail" id="myEmail">
<label for="myComments">Comments:</label>
<textarea name="myComments" id="myComments" rows="2"
cols="20"></textarea>
<input id="mySubmit" type="submit" value="Submit"> </form>
```

HTML5 Form Controls

- HTML5 offers a variety of new form controls for web developers.
 1. The **e-mail address input** form control is similar to the text box. Its purpose is to accept information that must be in e-mail format.

```
<label for="myEmail">E-mail:</label>
```

```
<input type="email" name="myEmail" id="myEmail">
```

2. The **URL input** form control is similar to the text box. It is intended to accept any valid type of URL or URI.

```
<label for="myWebsite">Suggest a Website:</label>
```

```
<input type="url" name="myWebsite" id="myWebsite">
```

Cont'd

3. The **telephone number input** form control is similar to the text box.

```
<label for="mobile">Mobile Number:</label>
```

```
<input type="tel" name="mobile" id="mobile">
```

4. The **search field** is similar to the text box and is used to accept a search term.

```
<label for="keyword">Search:</label>
```

```
<input type="search" name="keyword" id="keyword">
```

Cont'd

5. The **datalist form** control offers a convenient way to offer choices yet provide for flexibility on a form.

```
<label for="color">Favorite Color:</label>
```

```
<input type="text" name="color" id="color" list="colors">
```

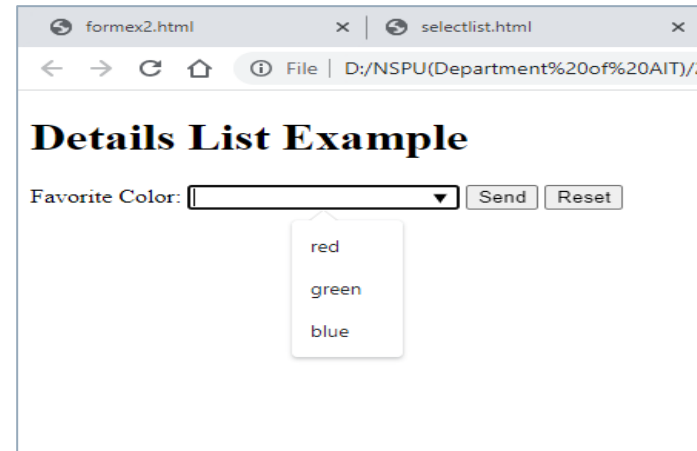
```
<datalist id="colors">
```

```
  <option value="red">
```

```
    <option value="green">
```

```
      <option value="blue">
```

```
</datalist>
```



Cont'd

6. The **slider form control** provides a visual, interactive user interface that accepts numerical information.

```
<label for="myChoice">Choose a number between 1 and 100:</label><br>
```

```
Low <input type="range" name="myChoice" id="myChoice" min="1"
```

```
max="100"> High
```

Additional attributes for slider, spinner, and date/time form controls

Attribute	Value
max	Maximum numeric value
min	Minimum numeric value
step	Incremental numeric step value

Cont'd

7. The **spinner form control** displays an interface that accepts numerical information and provides feedback to the user.

```
<label for="myChoice">Choose a number between 1 and 10:</label>
```

```
<input type="number" name="myChoice" id="myChoice" min="1" max="10">
```

8. HTML5 provides a variety of **calendar form controls** to accept date- and time-related information.

```
<label for="myDate">Choose a Date</label>
```

```
<input type="date" name="myDate" id="myDate">
```

Cont'd

7. The **spinner form control** displays an interface that accepts numerical information and provides feedback to the user.

```
<label for="myChoice">Choose a number between 1 and 10:</label>
```

```
<input type="number" name="myChoice" id="myChoice" min="1" max="10">
```

8. HTML5 provides a variety of **calendar form controls** to accept date- and time-related information.

```
<label for="myDate">Choose a Date</label>
```

```
<input type="date" name="myDate" id="myDate">
```

Cont'd

Date and time controls

Type	Attribute Value	Purpose	Format
date		A date	YYYY-MM-DD Example: January 2, 2016 is represented by "2016-01-02"
datetime		A date and time with time zone information; note that the time zone is indicated by the offset from UTC time	YYYY-MM-DDTHH:MM:SS-##:##Z Example: January 2, 2016, at exactly 9:58 a.m. Chicago time (CST) is represented by "2016-01-02T09:58:00-06:00Z"
datetime-local		A date and time without time zone information	YYYY-MM-DDTHH:MM:SS Example: January 2, 2016, at exactly 9:58 a.m. is represented by "2016-01-02T09:58:00"
time		A time without time zone information	HH:MM:SS Example: 1:34 p.m. is represented by "13:34:00"

Cont'd

Date and time controls

Type	Attribute Value	Purpose	Format
	month	A year and month	YYYY-MM Example: January 2016 is represented by "2016-01"
	week	A year and week	YYYY-W##, where ## represents the week in the year Example: The third week in 2016 is represented by "2016-W03"

Cont'd

9. The **color-well form control** displays an interface that offers a color-picker interface to the user.

```
<label for="myColor">Choose a color:</label>
```

```
<input type="color" name="myColor" id="myColor">
```

Thank You